

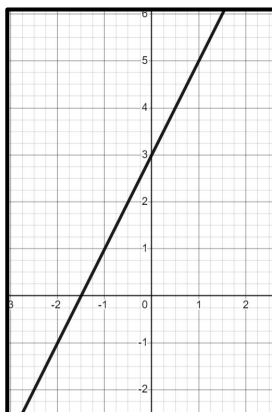


Finding the equation of the line $y = mx + c$

Task A: Writing the equation of a line from key features

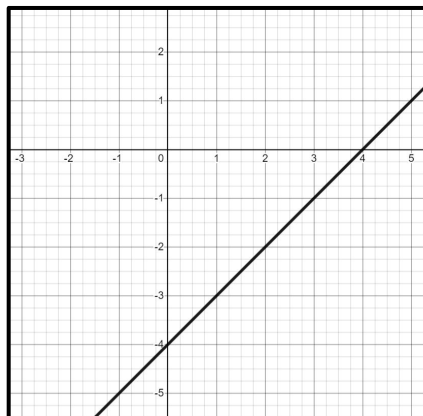
1) For each graph, work out an equation for the line in the form $y = mx + c$

a)



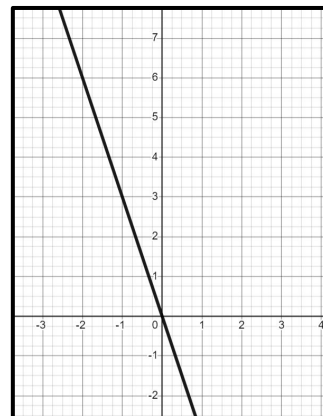
gradient: 2
y-intercept: $(0, 3)$
equation: $y = 2x + 3$

b)



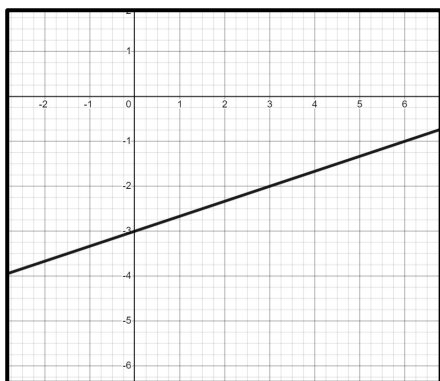
gradient: 1
y-intercept: $(0, -4)$
equation: $y = x - 4$

c)



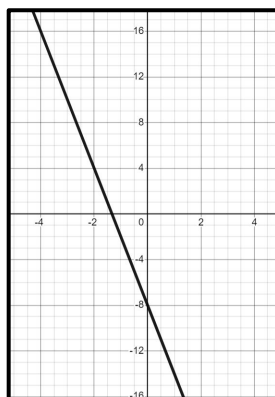
gradient: -3
y-intercept: $(0, 0)$
equation: $y = -3x$

d)



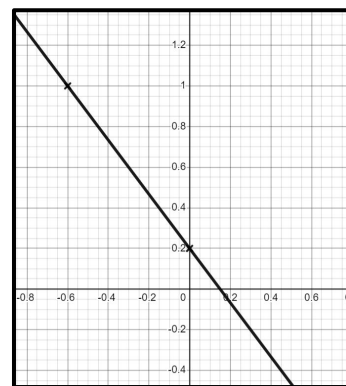
gradient: $\frac{1}{3}$
y-intercept: $(0, -3)$
equation: $y = \frac{1}{3}x - 3$

e)



gradient: -6
y-intercept: $(0, -8)$
equation: $y = -6x - 8$

f)



gradient: $-\frac{4}{3}$
y-intercept: $(0, 0.2)$
equation: $y = -\frac{4}{3}x + 0.2$



Task B: Finding the equation of a line

1) Calculate the equation of the line which has the table of values shown.

a)

x	0	1	2
y	-7	-3	1

 $y = 4x - 7$

b)

x	-1	0	1
y	14	3	-8

 $y = -11x + 3$

c)

x	0 ...	1	2	3
y	7 ...	5	3	1

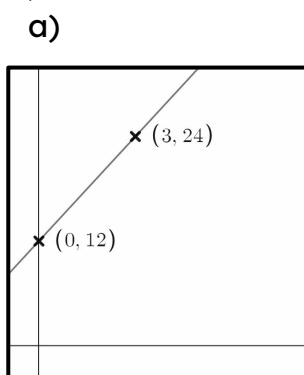
 $y = -2x + 7$

d)

x	0 ...	10	11	12
y	-22 ...	-12	-11	-10

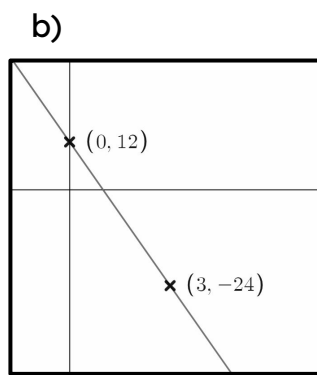
 $y = x - 22$

2) Calculate the equation of each line.



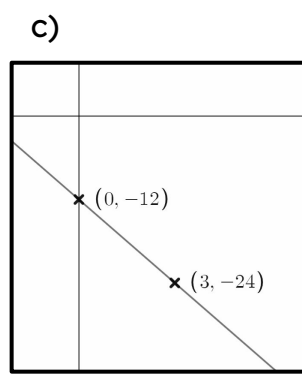
Change x	Change y
3	12
1	4

$$y = 4x + 12$$



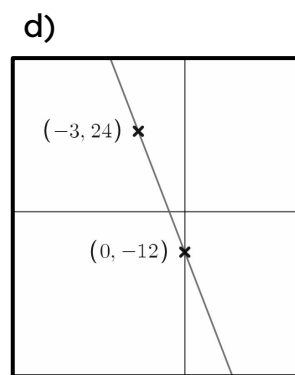
Change x	Change y
3	-36
1	-12

$$y = -12x + 12$$



Change x	Change y
3	-12
1	-4

$$y = -4x - 12$$



Change x	Change y
3	-36
1	-12

$$y = -12x - 12$$

3) Calculate the equation of the line passing through the coordinates given

a) (3, 5) and (5, 13) b) (3, 5) and (7, -15) c) (3, 5) and (-2, 10) d) (3, 5) and (-15, -1)

Change x	Change y
2	8
1	4

$$\text{Gradient} = 4$$

x	0 ...	3 ...	5
y	-7	5 ...	13

$$\text{Equation: } y = 4x - 7$$

Change x	Change y
4	-20
1	-5

$$\text{Gradient} = -5$$

x	0 ...	3 ...	7
y	20	5 ...	-9

$$\text{Equation: } y = -5x + 20$$

Change x	Change y
5	-5
1	-1

$$\text{Gradient} = -1$$

x	0	...	3
y	8	...	5

$$\text{Equation: } y = -x + 8$$

18	6
1	$\frac{1}{3}$

$$\text{Gradient} = \frac{1}{3}$$

x	0	...	3
y	4	...	5

$$\text{Equation: } y = \frac{1}{3}x + 4$$